ANALYTICAL PROBLEM SOLVING – ASSESSMENT

Name: Christina.y ***Register No:***1965260124

Course / Subject: Artificial Intelligence

1. Water Jug Problem

Given: A4-gallonjug and a 3-gallon jug. Neither has measuring marks.

Goal: Obtainexactly 2 gallons of water in the 4-gallon jug.

State representation:(amount in 4-gallon jug, amount in 3-gallon jug)

Step	*Action*	*State*
1	Fill the 3-gallon jug.	(0, 3)
2	Pour the 3-gallon jug into the 4-gallon jug.	(3, 0)
3	Fill the 3-gallon jug again.	(3, 3)
4	Pour into the 4-gallon jug until it is full.	(4, 2)
5	Empty the 4-gallon jug.	(0, 2)
6	Pour the remaining 2 gallons into the 4-gallon jug.	(2, 0)

Result: Exactly 2 gallons are obtained in the 4-gallon jug.

2. Mars Rover – Intelligent Agent

i.Percepts: Rocks, soil,obstacles, temperature, atmospheric conditions, distance, terrain conditions, sample locations, and signs of water or minerals.

ii. Operating environment: Partially observable, dynamic, sequential, uncertain, continuous, and mainly single-agent.

iii. Actions: Move, turn, collect and analyze samples, take photographs, avoid obstacles, navigate, and communicate with Earth.

iv. Performance measures: Useful samples collected, accuracy of analysis, distance travelled, time taken, energy consumption, obstacle avoidance, scientific information obtained, and mission completion.

v. Suitable architecture: Model-Based Utility-Based Agent. It maintains information about the environment and chooses useful actions while considering goals such as sample collection, safety, and energy conservation.

3. 8-Queens Problem

Problem:Place 8 queens on an 8 × 8 chessboard so that no two queens attack each other.

Initial state: Empty chessboard.

State: A partial arrangement of queens on the chessboard.

Actions:Place a queen in a safe position in the next column.

Transition model: Adding a queen to a valid position produces a new state.

Goal test: All 8 queens are placed with no two queens in the same row, column, or diagonal.

Path cost: Each queen placement has a cost of 1; a complete solution has cost 8.

Suitable search: Backtracking / Depth First Search is commonly used because invalid arrangements can be rejected immediately.

One valid solution: [1, 5, 8, 6, 3, 7, 2, 4], where each number represents the row of the queen in successive columns.

4. OLA Cab Booking – Problem-Solving Agent

Problem: A customerwantsto travel from a source to a destination using an OLA cab booking application. The customer may select Mini, Micro, Sedan, Prime, or Shared based on cost, comfort, and travel requirements.

Agent type: Utility-Based Agent.

Reason: The agent compares available choices using factors such as fare, travel time, comfort, and customer preference, then selects the option with the highest utility.

Pseudocode:

START
Input source location
Input destination
location Find available
cabs

For each available
cab: Calculate fare
Calculate travel time
Check comfort level
Calculate utility

Select cab with highest utility
Display selected cab, fare and estimated travel
time
Ask customer for
confirmation If customer
confirms:
Book the cab
Display booking details
Else:
Cancel booking
STOP

5. Logistics Delivery Network – Uniform Cost Search

Given edge costs: →SA=1,S→G = 12, A→B = 3, A→C = 1, B→D = 3, C→D = 1, C→G = 2, D→G = 3. UCS

Uniform Cost Search (UCS): always expands the path having the lowest total cost so far.

Stage	*Path / Node*	*Cost*
Start	S	0
1	S → A	1
2	S → A → C	2
3	S → A → C → D	3
4	S → A → C →G	4

Comparison: S → A →C→Ghascost 1 + 1 + 2 = 4, while the direct path S → G costs 12. Therefore, UCS selects the lower-cost route.

Final Answer: Least-cost path = S →A →C →G; Minimum cost = 4.